What is Future Makers?

How could technology help close the gender gap in STEM?

Over two days, your team will identify a problem or barrier contributing to the gender gap in STEM and use technology to create an innovative solution.

There is deliberately no prescribed technology or type of solution.

You could:

- build a game
- create a web or mobile app
- prototype an AI solution
- analyse data and visualise what you discover
- develop an interactive experience
- create something completely different

We want you to choose the technology that makes the most sense for your idea and your team's skills.

What could we build?

These aren't challenges you have to choose from. They're just examples to get you thinking.

🎮 Game development

Problem: Young people may have limited awareness of the range of careers available in technology.

Idea: Create an interactive game where players explore different STEM careers, complete challenges and discover the skills involved.

🤖 AI

Problem: Someone interested in STEM might not know which career would suit them or how to get started.

Idea: Prototype an AI career coach that lets someone explore STEM careers based on their interests and shows possible routes into them.

📱 App / Web

Problem: People interested in STEM may struggle to find relatable role models or support.

Idea: Create an experience that connects students with STEM role models, mentors, communities, opportunities or career stories.

📊 Data

Problem: It's difficult to tackle inequality if you can't clearly see where the gaps exist.

Idea: Analyse a STEM dataset and build an interactive dashboard highlighting differences across subjects, education or careers, then suggest where interventions could have the biggest impact.

🕹️ Interactive learning

Problem: Stereotypes around who “belongs” in technology can start early.

Idea: Create a short interactive experience that challenges common STEM stereotypes and introduces users to the variety of people, skills and careers found in technology.

🔐 Cyber

Problem: Cyber security can sometimes feel inaccessible to people who've never experienced it.

Idea: Build an accessible cyber challenge or interactive experience that introduces security concepts while showcasing different cyber careers.

Start by asking yourself “What is a real barrier contributing to the gender gap in STEM, and how could technology help?” A great idea doesn't need to be the most technically complicated one. Think about what that solution could be, but also consider the skills within your team to make that a reality.

Think about: Problem → Audience → Idea → Technology → Impact

For example: Girls aren't considering cyber because they don't understand what cyber careers actually involve → Secondary-school students → Interactive cyber career experience → Web/game technology → Makes cyber careers more understandable and relatable

How will the challenge work?

Day 1: Wednesday 4 November

10:00–16:00

This is your opportunity to:

- Understand the problem
- Decide who you're designing for
- Explore ideas
- Choose your solution
- Divide responsibilities across the team
- Start creating your prototype

Day 2: Thursday 5 November

10:00–16:00

Continue developing and refining your idea.

Your Lloyds Banking Group helpers will work with you during the event, and your solution will be assessed against the judging criteria.

The top three teams will then present their solution to the judging panel.

Final pitch

Each finalist will have 10 minutes. Don't spend two days building something brilliant and then forget to explain why it matters.

Your pitch should make it easy to understand:

1. What problem did you identify? What aspect of the gender gap are you tackling?

2. Who are you helping? Who is your target audience?

3. What did you create? Show us your solution.

4. Why did you choose this technology? How does it support the problem you're solving?

5. What impact could it have? How could your idea genuinely help?

6. What did you learn? What changed or evolved during the challenge?

We're not judging who can build the biggest thing in two days. We're looking for teams who understand the problem, use technology thoughtfully and show us an idea that could make a meaningful difference to the gender gap in STEM.

Frequently Asked Questions

Do we have to build an app?

No. Your solution could be an app, game, AI prototype, data analysis, website, interactive experience or something completely different.

Do we have to use AI?

No. AI is just one option. Choose the technology that works best for your problem and your team.

Do we have to build a game?

No. Games are one possible approach, not the challenge itself. This year's brief was deliberately broadened beyond games.

Does our prototype have to be completely finished?

No. Treat it as a prototype. We'd rather understand a strong idea and see enough to demonstrate how it could work than have you spend all your time trying to build every possible feature. During your presentation you could always include what else you would have done if you had more time.

Can we use technology or tools we already know?

Yes. You're free to choose your preferred technologies and use your own devices to create your solution.

Can we use something we've never used before?

Yes. Experimenting and learning is very much part of the spirit of the challenge.

Do we all need to be programmers?

No. Teams are encouraged to combine different skills, including software development, data science, design, analysis, presentation and product thinking.

How many people are in a team?

Teams are made up of up to six students. We encourage teams to have a mixture of skills rather than six people with the same expertise, and ask that the majority of the team are female/non-binary.

Will there be people to help us?

Yes. Lloyds Banking Group helpers will support teams throughout the two days. Think of them as people you can bounce ideas off, rather than people who will build the solution for you.

Is the most technically impressive solution going to win?

Not necessarily. Technical execution matters, but don't forget the bigger challenge: does your solution actually help address the gender gap in STEM? Your story, problem, audience, idea and potential impact matter too.

Should we spend most of our time coding?

No. Building is only part of the challenge. Make sure you leave enough time to think about the problem, test

Frequently Asked Questions

Who presents?

The top three teams will be selected to present their solution to the judging panel. You can choose within your team if you'd all like to present.

How long is the final presentation?

10 minutes per team.

When will the winners be announced?

Teams are invited to the Ada Scotland Festival Closing Celebration from 16:00–18:00 on Thursday 5 November, where the competition winners will be announced.

What should we bring?

Bring a personal laptop with Wi-Fi capability and your power cable. If possible, having HDMI or USB-C connectivity will also help with collaboration and presenting. Download any tools you'd like to use beforehand.

What exactly do you mean by "closing the gender gap in STEM"?

We want you to explore a barrier that contributes to gender inequality in STEM and think about how technology could help. This could relate to education, confidence, awareness, stereotypes, career choices, progression, role models, opportunities or something else you identify.

Do we need to tackle the whole gender gap or all of STEM?

No. In fact, we'd encourage you to focus. You could tackle one specific problem within areas such as technology, engineering, cyber, data, gaming or another STEM field.

Who should our solution be aimed at?

That's up to you. You might focus on school pupils, university students, graduates, people considering a career change, people already working in STEM, employers or another audience. Just be clear about who you're designing for and why.

What counts as a prototype? Does it need to be fully working?

No, it doesn't need to be a finished product. Your prototype simply needs to demonstrate your idea effectively. That could be a working application, a small game, a clickable prototype, an AI proof of concept, data visualisation or another way of bringing your idea to life.

Do we have to code something?

No. Technology should support your idea rather than dictate it. The important thing is that you're able to demonstrate your proposed solution and explain how it could make a difference.

What technology are we allowed to use?

You can choose the technology that best suits your idea and skills. That could include AI, data analysis, web or mobile development, game development, design/prototyping tools or something completely different. The challenge was intentionally designed to allow teams to choose their own technology.

Can we use AI or existing tools, libraries and resources?

Yes, you can use tools and resources to help you create your prototype. If existing technology forms an important part of your solution, we'd encourage you to explain what your team created and how you've used the technology.

What will we be judged on?

You'll be assessed against the judging criteria we're providing for the challenge. Remember, it isn't just about what you build. Think about the problem you're solving, who it helps, the thinking behind your solution, its potential impact and how clearly you communicate your idea.

What can our Lloyds Banking Group helpers help us with?

They're there to support you throughout the challenge, challenge your thinking and give you another perspective on your idea. On Day 2, they'll also score your team against the judging criteria to help determine the three finalists.

What should we include in our 10-minute final presentation?

Tell us a story: What problem did you discover? Who does it affect? What did you create? Why did you choose this approach? How does it work? And most importantly, how could it help close the gender gap in STEM? The top three teams will present to the final judging panel.